

三. 微分中值定理及导数的应用

1. 微分中值定理

定义 设 $f(x)$ 在 x_0 的某个邻域 $(x_0 - \delta, x_0 + \delta)$ 内, 满足 $\forall x \in (x_0 - \delta, x_0 + \delta)$, 恒有 $f(x_0) \geq f(x)$, 则称 x_0 为 $f(x)$ 的一个极大值点, $f(x_0)$ 为 $f(x)$ 的极大值. 类似地, $f(x_0) \leq f(x)$, 则称 x_0 为 $f(x)$ 的一个极小值点, $f(x_0)$ 为 $f(x)$ 的极小值.

费马引理 若 $f(x)$ 在极值点 x_0 处可导, 则 $f'(x_0) = 0$.

罗尔中值定理

设 $f(x)$ 在 $[a, b]$ 上满足

- (1) $f(x)$ 在 $[a, b]$ 上连续;
- (2) $f(x)$ 在 (a, b) 上可导;
- (3) $f(a) = f(b)$.

则 $\exists \xi \in (a, b)$, 使 $f'(\xi) = 0$.

例1: 已知 $a_n + \frac{n}{n-1}a_{n-1} + \frac{n}{n-2}a_{n-2} + \cdots + \frac{n}{2}a_2 + na_1 = 0$,

证明 $a_n x^{n-1} + a_{n-1} x^{n-2} + \cdots + a_1 = 0$ 在 $(0, 1)$ 有根.

证: 令 $f(x) = \frac{a_n}{n} x^n + \frac{a_{n-1}}{n-1} x^{n-1} + \cdots + \frac{a_2}{2} x^2 + a_1 x$

则 $f(x) \in C[0, 1]$, $f(x) \in D(0, 1)$.

且 $f'(x) = a_n x^{n-1} + a_{n-1} x^{n-2} + \cdots + a_2 x + a_1$

$f(0) = 0$, $f(1) = \frac{a_n}{n} + \frac{a_{n-1}}{n-1} + \cdots + \frac{a_2}{2} + a_1 = \frac{1}{n}(a_n + \frac{n}{n-1}a_{n-1} + \cdots + na_1) = 0$

即 $f(0) = f(1)$

由罗尔中值定理, $\exists \xi \in (0, 1)$ 使 $f'(\xi) = 0$

即 $a_n x^{n-1} + a_{n-1} x^{n-2} + \cdots + a_1 = 0$ 在 $(0, 1)$ 有根.

例: 设 $f(x) \in C[a, b]$, $f(x) \in D(a, b)$, $f(a) = f(b) = 0$.

证明: $\exists \xi \in (a, b)$, 使 $f'(\xi) + 2\xi f(\xi) = 0$.

证. 令 $F(x) = f(x) \cdot e^{x^2}$, $F(x) \in C[a, b]$, $F(x) \in D(a, b)$

$$F'(x) = [f'(x) + 2x f(x)] e^{x^2}, \quad F(a) = f(a) e^{a^2} = 0, \quad F(b) = f(b) e^{b^2} = 0.$$

由罗尔中值定理,

$\exists \xi \in (a, b)$, 使 $F'(\xi) = [f'(\xi) + 2\xi f(\xi)] e^{\xi^2} = 0$,

即 $f'(\xi) + 2\xi f(\xi) = 0$.

$$[f(x) e^{g(x)}]' = e^{g(x)} [f'(x) + f(x) g'(x)]$$

例 设 $f: [0, 1] \rightarrow [0, 1]$ 上连续, f 在 $(0, 1)$ 上可导且 $0 < f(x) < 1, x \in (0, 1)$.

证明 $\exists \xi \in (0, 1)$, 使 $f'(\xi) + \frac{1}{\xi} f(\xi) = 2$.

证. 令 $F(x) = x f(x) - x^2$, $F(x) \in C[a, b]$, $F \in D(a, b)$, $F'(x) = x f'(x) + f(x) - 2x$
 $F(0) = 0$, $F(1) = f(1) - 1 < 0$, $F(x) = x(f(x) - x)$.

令 $g(x) = f(x) - x$, $g(x) \in C[0, 1]$, $g(0) = f(0) > 0$, $g(1) = f(1) - 1 < 0$

$\exists a \in (0, 1)$, 使 $g(a) = 0$, 即 $F(a) = 0$. $F(a) = F(0)$

由罗尔中值定理,

$\exists \xi \in (0, a)$, 使 $F'(\xi) = \xi [f'(\xi) + \frac{1}{\xi} f(\xi) - 2] = 0$.

即 $\exists \xi \in (0, 1)$, 使 $f'(\xi) + \frac{1}{\xi} f(\xi) = 2$.

拉格朗日中值定理

设 $f(x)$ 在 $[a, b]$ 上满足

1) $f(x)$ 在 $[a, b]$ 上连续;

2) $f(x)$ 在 (a, b) 上可导;

则 $\exists \xi \in (a, b)$, $f'(\xi) = \frac{f(b) - f(a)}{b - a}$.

称 $f(x_0 + \Delta x) - f(x_0) = f'(\xi) \Delta x$
为有限增量公式

推论1 若 $f(x)$ 在 $[a, b]$ 上连续, 在 (a, b) 上可导, 且 $f'(x) = 0$, 则 $f(x) = C$ (C 为常数).

推论2 设 $f(x), g(x) \in C[a, b]$, $f(x), g(x) \in D(a, b)$, 则 $f'(x) = g'(x)$ 的必要条件为对于 $\forall x \in (a, b)$, $\exists$ 常数 C , 使 $f(x) = g(x) + C$.

例: 证明: $\arctan x + \operatorname{arccot} x = \frac{\pi}{2}$, $x \in \mathbb{R}$.

令 $f(x) = \arctan x + \operatorname{arccot} x$, $x \in \mathbb{R}$

$$f'(x) = \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0, \text{ 故 } f(x) = C, C \text{ 为常数}$$

$$f(x) = f(1) = \frac{\pi}{2}, \text{ 即 } \arctan x + \operatorname{arccot} x = \frac{\pi}{2}.$$

例: 设 $f(x) \in C[0, 1]$, $f(x) \in D(0, 1)$, $f(0) = 0$, $f(1) = 0$,

证明 $\exists 0 < \xi < \eta < 1$, 使 $\frac{1}{f'(\xi)} + \frac{1}{f'(\eta)} = 2$

证. 由于 $f(x) \in C[0, 1]$, $f(0) < \frac{1}{2} < f(1)$.

故 $\exists a \in (0, 1)$, 使 $f(a) = \frac{1}{2}$.

$f(x) \in C[0, a]$, $f(x) \in D(0, a)$

且 $f(x) \in C[a, 1]$, $f(x) \in D(a, 1)$,

由拉格朗日中值定理,

$$\exists \xi \in (0, a), \text{ 使 } f'(\xi) = \frac{f(a) - f(0)}{a - 0} = \frac{1}{2a}$$

$$\exists \eta \in (a, 1), \text{ 使 } f'(\eta) = \frac{f(1) - f(a)}{1 - a} = \frac{1}{2-2a}$$

$$\text{从而 } f'(\xi) + f'(\eta) = 2a + 2-2a = 2.$$

分析: 即寻找 a , 使 $\exists 0 < \xi < a < \eta < 1$.

$$\text{使 } f'(\xi) = \frac{f(a) - f(0)}{a - 0}, \quad f'(\eta) = \frac{f(1) - f(a)}{1 - a}$$

$$\frac{1}{f'(\xi)} + \frac{1}{f'(\eta)} = \frac{a}{f(a)} + \frac{1-a}{1-f(a)} = 2.$$

$$2f(a) - (2a+1)f(a) + a = 0$$

$$[f(a) - a][2f(a) - 1] = 0$$

$$f(a) = a \text{ 不具有任意性, 故 } f(a) = \frac{1}{2}$$

$$\text{而 } f(x) \in C[0, 1], \quad f(0) < \frac{1}{2} < f(1)$$

$$\text{故 } \exists a \in (0, 1), \text{ 使 } f(a) = \frac{1}{2}.$$

例: 设 $f(x) \in C[0, 1]$, $f(x) \in D(0, 1)$, $f(0) = f(1)$.

证明 $\exists 0 < \xi < \eta < 1$, 使 $f'(\xi) + 3f'(\eta) = 0$.

一. (拉格朗日中值定理)

$$f(x) \in C[0, 1], f(x) \in D(0, 1)$$

$$\exists \xi \in (0, \frac{1}{4}), f'(\xi) = \frac{f(\frac{1}{4}) - f(0)}{\frac{1}{4} - 0}$$

$$\exists \eta \in (\frac{1}{4}, 1), f'(\eta) = \frac{f(1) - f(\frac{1}{4})}{1 - \frac{1}{4}}$$

$$f'(\xi) + 3f'(\eta) = 4f(\frac{1}{4}) - 4f(0) + 3[\frac{4}{3}f(1) - \frac{4}{3}f(\frac{1}{4})] = 0.$$

二. (罗尔中值定理)

$$\text{令 } F(x) = f(x) - f(1-3x)$$

$$F(x) \in C[0, \frac{1}{4}], F \in D(0, \frac{1}{4})$$

$$F'(x) = f'(x) + 3f'(1-3x),$$

$$F(0) = f(0) - f(1) = 0, F(\frac{1}{4}) = f(\frac{1}{4}) - f(\frac{1}{4}) = 0.$$

$$\text{由罗尔中值定理, } \exists \xi \in (0, \frac{1}{4}), \text{使 } F'(\xi) = 0$$

$$\text{即 } f'(\xi) + 3f'(1-3\xi) = 0. \text{ 取 } \eta = 1-3\xi, \text{ 则 } 0 < \xi < \frac{1}{4} < \eta < 1.$$

$$\text{且 } f'(\xi) + 3f'(\eta) = 0.$$

柯西中值定理

设 $f(x)$ 和 $g(x)$ 在 $[a, b]$ 上满足

(1) 在 $[a, b]$ 上连续;

(2) 在 (a, b) 上可导, 且 $g'(x) \neq 0$;

$$\text{则 } \exists \xi \in (a, b), \text{ 使 } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

例. 证明: $\exists \xi \in (1, e)$, 使 $\sin 1 = \cos \ln \xi$ 成立.

一. (柯西中值定理)

分析: 取 $0 < \xi < a < \eta < 1$.

$$f'(\xi) + 3f'(\eta) = \frac{f(a) - f(0)}{a - 0} + 3 \cdot \frac{f(1) - f(a)}{1 - a} = 0$$

$$f(a) - f(0) - a f(a) + a f(0) + 3a f(1) - 3a f(a) = 0$$

$$(1-4a)f(a) + (4a-1)f(0) = 0$$

$$[f(0) - f(a)](1-4a) = 0, \text{ 即取 } a = \frac{1}{4}.$$

令 $F(x) = \sin \ln x$, $f(x) = \ln x$, $F'(x) = \frac{1}{x} \cos \ln x$, $f'(x) = \frac{1}{x}$
 $F(x), f(x) \in C[1, e]$, $F(x), f(x) \in D(1, e)$, $f'(x) \neq 0$.

由柯西中值定理,

$$\exists \xi \in (1, e), \frac{F(e) - F(1)}{f(e) - f(1)} = \frac{F'(\xi)}{f'(\xi)}$$

$$\frac{F(e) - F(1)}{f(e) - f(1)} = \frac{\sin 1 - 0}{1 - 0} = \sin 1, \quad \frac{F'(\xi)}{f'(\xi)} = \cos \ln \xi,$$

$$\text{即 } \exists \xi \in (1, e), \sin 1 = \cos \ln \xi.$$

二. (罗尔中值定理)

令 $F(x) = \sin \ln x - \sin 1 \cdot \ln x$, $F'(x) = \frac{1}{x} \cos \ln x - \frac{1}{x} \sin 1$

$F(x) \in C[1, e]$, $F(x) \in D(1, e)$, $F(1) = F(e) = 0$,

由罗尔中值定理,

$$\exists \xi \in (1, e), F'(\xi) = \frac{1}{\xi} \cos \ln \xi - \frac{1}{\xi} \sin 1 = 0, \text{ 即 } \sin 1 = \cos \ln \xi.$$

2. 洛必达法则

定理 若 (1) $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} F(x) = 0$.

(2) $x \in \dot{U}_\delta(a)$, $f(x), F(x)$ 均可导且 $F'(x) \neq 0$,

(2) $\lim_{x \rightarrow a} \frac{f'(x)}{F'(x)}$ 存在或为 ∞ .

$$\text{则 } \lim_{x \rightarrow a} \frac{f(x)}{F(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{F'(x)}.$$

定理 若 (1) $\lim_{x \rightarrow a} |f(x)| = \lim_{x \rightarrow a} |F(x)| = \infty$

(2) $x \in \dot{U}_\delta(a)$, $f(x), F(x)$ 均可导且 $F'(x) \neq 0$,

(2) $\lim_{x \rightarrow a} \frac{f'(x)}{F'(x)}$ 存在或为 ∞ .

则 $\lim_{x \rightarrow a} \frac{f(x)}{F(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{F'(x)}$.

例: 求 $\lim_{x \rightarrow 0^+} x^{x-1}$

$$\begin{aligned} \lim_{x \rightarrow 0^+} x^{x-1} &= \lim_{x \rightarrow 0^+} x^{e^{\ln x} - 1} = \lim_{x \rightarrow 0^+} e^{(e^{\ln x} - 1) \ln x} \\ &= e^{\lim_{x \rightarrow 0^+} (e^{\ln x} - 1) \ln x} = e^{\lim_{x \rightarrow 0^+} x \ln^2 x} \\ &= e^{\lim_{x \rightarrow 0^+} \left(\frac{\ln x}{\frac{1}{x}}\right)^2} = e^{\lim_{x \rightarrow 0^+} \left(\frac{\frac{1}{x}}{\frac{1}{2x}}\right)^2} = e^{\lim_{x \rightarrow 0^+} \frac{4}{x}} = e^\infty = +\infty \end{aligned}$$

例: 求 $\lim_{x \rightarrow +\infty} \left(\frac{3^x - 1}{2x}\right)^{\frac{1}{x}}$

$$\begin{aligned} \lim_{x \rightarrow +\infty} \left(\frac{3^x - 1}{2x}\right)^{\frac{1}{x}} &= \lim_{x \rightarrow +\infty} e^{\frac{\ln(3^x - 1) - \ln(2x)}{x}} = e^{\lim_{x \rightarrow +\infty} \frac{\ln(3^x - 1)}{x} - \frac{\ln(2x)}{x}} \\ &= e^{\lim_{x \rightarrow +\infty} \frac{\frac{1}{3^x - 1} \cdot 3^x \ln 3}{1} - \lim_{x \rightarrow +\infty} \frac{\frac{1}{2x} \cdot 2}{1}} \\ &= e^{\lim_{x \rightarrow +\infty} \frac{\ln 3}{1 - \frac{1}{3^x}} - \lim_{x \rightarrow +\infty} \frac{1}{x}} \\ &= e^{\ln 3} = 3 \end{aligned}$$

例: 求 $\lim_{x \rightarrow \infty} \left[\frac{(1 + \frac{1}{x})^x}{e}\right]^x$

$$\begin{aligned} \lim_{x \rightarrow \infty} \left[\frac{(1 + \frac{1}{x})^x}{e}\right]^x &= \lim_{x \rightarrow \infty} \left[\frac{e^{x \ln(1 + \frac{1}{x})}}{e}\right]^x = \lim_{x \rightarrow \infty} e^{[x \ln(1 + \frac{1}{x}) - 1]x} \\ &= e^{\lim_{x \rightarrow \infty} \frac{\ln(1 + \frac{1}{x}) - \frac{1}{x}}{(\frac{1}{x})^2}} \stackrel{t \leftrightarrow \frac{1}{x}}{=} e^{\lim_{t \rightarrow 0} \frac{\ln(1+t) - t}{t^2}} \\ &= e^{\lim_{t \rightarrow 0} \frac{\frac{1}{1+t} - 1}{2t}} = e^{\lim_{t \rightarrow 0} -\frac{1}{2(1+t)}} = e^{-\frac{1}{2}} \end{aligned}$$

例1: 求 $\lim_{x \rightarrow 0} \frac{\ln(1+\sin^2 x) - \sin^2(\sin x)}{x^4}$

$$\lim_{x \rightarrow 0} \frac{\ln(1+\sin^2 x) - \sin^2(\sin x)}{x^4} = \lim_{x \rightarrow 0} \frac{\ln(1+\sin^2 x) - \sin^2(\sin x)}{\sin^4 x}$$

$$\xrightarrow{t \rightarrow \sin x} \lim_{t \rightarrow 0} \frac{\ln(1+t^2) - \sin^2 t}{t^4} = \lim_{t \rightarrow 0} \frac{\ln(1+t^2) - t^2 + t^2 - \sin^2 t}{t^4}$$

$$= \lim_{t \rightarrow 0} \frac{\ln(1+t^2) - t^2}{t^4} + \lim_{t \rightarrow 0} \frac{(t+\sin t)(t-\sin t)}{t^4}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{1}{1+t^2} \cdot 2t - 2t}{4t^3} + \lim_{t \rightarrow 0} \frac{(t+\sin t) \cdot \frac{1}{6} t^3}{t^4}$$

$$= \lim_{t \rightarrow 0} -\frac{1}{2(1+t^2)} + \lim_{t \rightarrow 0} \frac{1}{6} \frac{t+\sin t}{t}$$

$$= -\frac{1}{2} + \frac{1}{6} \lim_{t \rightarrow 0} \left(1 + \frac{\sin t}{t}\right) = -\frac{1}{2} + \frac{1}{6} \times 2 = -\frac{1}{6}$$

定理 (Stolz 定理)

设数列 $\{a_n\}, \{b_n\}$, $b_n > 0$ 且 b_n 递增, $\lim_{n \rightarrow \infty} b_n = +\infty$, 则

若 $\lim_{n \rightarrow \infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A$ (或 ∞), 则 $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = A$ (或 ∞).

例: 设 $a_{n+1} = \sin a_n$, $a_1 \in [0, \pi)$, 求 $\lim_{n \rightarrow \infty} \sqrt[n]{a_n}$.

$a_{n+1} = \sin a_n < a_n$, $\{a_n\}$ 递减, $a_n \in [0, 1]$ 有界, 故可设 $\lim_{n \rightarrow \infty} a_n = a$

$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \sin a_n$, 即 $a = \sin a$, $a \in [0, \pi)$, $a = 0$.

即 $\lim_{n \rightarrow \infty} a_n = 0$.

$$\lim_{n \rightarrow \infty} \frac{1}{n a_n^2} = \lim_{n \rightarrow \infty} \frac{\frac{1}{a_n^2}}{n} \stackrel{\text{Stolz}}{=} \lim_{n \rightarrow \infty} \frac{\frac{1}{a_{n+1}^2} - \frac{1}{a_n^2}}{(n+1) - n} = \lim_{n \rightarrow \infty} \frac{1}{\sin^2 a_n} - \frac{1}{a_n^2}$$

$$\stackrel{a_n \leftrightarrow t}{=} \lim_{t \rightarrow 0} \frac{1}{\sin^2 t} - \frac{1}{t^2} = \lim_{t \rightarrow 0} \frac{t^2 - \sin^2 t}{t^2 \sin^2 t} = \lim_{t \rightarrow 0} \frac{t^2 - \sin^2 t}{t^4}$$

$$= \lim_{t \rightarrow 0} \frac{(t + \sin t)(t - \sin t)}{t^4} = \lim_{t \rightarrow 0} \frac{(t + \sin t) \cdot \frac{1}{6} t^3}{t^4} = \frac{1}{3}$$

从而 $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \sqrt[3]{3}$.

定理 (达布定理)

若 $f(x) \in D[a, b]$, $\exists x_1, x_2 \in [a, b]$, f' 介于 $f'(x_1)$ 与 $f'(x_2)$ 之间, 则 $\exists \xi$ 介于 x_1 与 x_2 之间, 使 $f'(\xi) = f'(\eta)$.

3 泰勒公式

定义 设 $f(x)$ 在 x_0 处有 n 阶导数,

$$P_n(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

称 $P_n(x)$ 为 $f(x)$ 在 x_0 处的泰勒多项式.

记 $R_n(x) = f(x) - P_n(x)$, 称 R_n 为 $f(x)$ 的 n 阶泰勒多项式的余项.

即 $f(x) = P_n(x) + R_n(x)$.

定理 设 $f(x)$ 在 x_0 处有 n 阶导数, 在 x_0 的一个邻域内, 有

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + R_n(x),$$

其中 $R_n(x) = o(|x-x_0|^n)$, 称之为带 Peano 余项的 n 阶泰勒公式.

当 $x_0 = 0$, $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + o(x^n)$
称之为带 Peano 余项的麦克劳林公式。

引理: 设 $f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \dots + a_n(x-x_0)^n + o((x-x_0)^n)$
且 $f(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)^2 + \dots + b_n(x-x_0)^n + o((x-x_0)^n)$
则 $a_i = b_i$, $i = 0, 1, \dots, n$.

注: $f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \dots + a_n(x-x_0)^n + o((x-x_0)^n)$

$\Rightarrow a_k = \frac{f^{(k)}(x_0)}{k!}$, 若 $f(x)$ 在 x_0 处连续, $a_0 = \frac{f(x_0)}{1!}$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \frac{e^{\theta x}}{(n+1)!}x^{n+1};$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{\cos \theta x}{(2n+1)!}x^{2n+1};$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + (-1)^{n+1} \frac{\cos \theta x}{(2n+2)!}x^{2n+2};$$

$$(1+x)^\mu = 1 + \mu x + \frac{\mu(\mu-1)}{2!}x^2 + \dots + \frac{\mu(\mu-1)\dots(\mu-n+1)}{n!}x^n$$

$$+ \frac{\mu(\mu-1)\dots(\mu-n)}{(n+1)!}(1+\theta x)^{\mu-n-1}x^{n+1};$$

$$\ln(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!} - \dots + (-1)^{n-1} \frac{x^n}{n!} + (-1)^n \frac{x^{n+1}}{(n+1)(1+\theta x)^{n+1}}.$$

例: 设 $f(x) = x^2 + x^3 \sin x e^{\tan x} \frac{1}{2+x^2}$, 求 $f^{(4)}(0)$, $f^{(5)}(0)$.

$$x \rightarrow 0, f(x) = x^2 + \frac{1}{2}x^5 + o(x^5)$$

$$\frac{f^{(4)}(0)}{4!} = 0, \frac{f^{(5)}(0)}{5!} = \frac{1}{2}, f^{(4)}(0) = 0, f^{(5)}(0) = 60.$$

例: 设 $f(x)$ 在 $\mathbb{R}$ 上有二阶导数, 且 $\lim_{x \rightarrow 0} \left[\frac{f(x) + e^x + \ln(1+2x)}{x} \right]^{\frac{1}{x}} = e^2$,
求 $f(0)$, $f'(0)$, $f''(0)$.

$$\lim_{x \rightarrow 0} \left[\frac{f(x) + e^x + \ln(1+2x)}{x} \right]^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \frac{\ln[f(x) + e^x + \ln(1+2x)]}{x^2}} = e^2$$

$$\lim_{x \rightarrow 0} \frac{\ln[f(x) + e^x + \ln(1+2x)]}{x^2} = 2,$$

$$\text{从而} \lim_{x \rightarrow 0} \frac{f(x) + e^x + \ln(1+2x) - 1}{x^2} = 2.$$

$$\text{故 } f(x) + e^x + \ln(1+2x) - 1 = 2x^2 + o(x^2)$$

$$\begin{aligned} f(x) &= -e^x - \ln(1+2x) + x + 2x^2 + o(x^2) \\ &= -(1+x+\frac{1}{2}x^2+o(x^2)) - [2x - \frac{(2x)^2}{2} + o(x^2)] + x + 2x^2 + o(x^2) \\ &= -1 - 2x + \frac{1}{2}x^2 + o(x^2) \end{aligned}$$

$$\text{则 } f(0) = -1, \frac{f'(0)}{1!} = -2, \frac{f''(0)}{2!} = \frac{1}{2}$$

$$f'(0) = -2, f''(0) = 1.$$

定理 (泰勒中值定理)

设 $f(x)$ 在 $[a, b]$ 上连续, $x_0 \in [a, b]$, $f^{(n)}(x_0)$ 连续, $f(x)$ 在 (a, b) 上除 x_0 以外均有 $n+1$ 阶导数, 则 $\forall x \in [a, b]$ 且 $x \neq x_0$ 时 $\exists \xi$ 介于 x 与 x_0 之间, 使

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$$

称 $R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$ 为拉格朗日余项.

$$\text{记 } \theta = \frac{\xi - x_0}{x - x_0} \in (0, 1), \quad \xi = x_0 + \theta(x - x_0) = x_0 + \theta \Delta x$$

$$\text{则 } f(x + \Delta x) = f(x_0) + \frac{f'(x_0)}{1!}\Delta x + \cdots + \frac{f^{(n)}(x_0)}{n!}\Delta x^n + \frac{f^{(n+1)}(x_0 + \theta \Delta x)}{(n+1)!}\Delta x^{n+1}$$

$$\text{如 } \frac{1}{1+x} = 1 - x + x^2 - \dots + (-1)^n x^n + (-1)^{n+1} \frac{1}{(1+\theta x)^{n+2}} x^{n+1}.$$

例. 设 $f(x)$ 在 $[-1, 1]$ 上三次可微, 证明: $\exists \xi \in [-1, 1]$,

$$\text{使 } \frac{f'''(\xi)}{6} = \frac{f(1) - f(-1)}{2} - f'(0).$$

证. 由题意, $\exists \xi \in [-1, 1]$, 使

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{6}f'''(\xi)x^3$$

$$f(1) = f(0) + f'(0) + \frac{1}{2}f''(0) + \frac{1}{6}f'''(\xi_1)$$

$$f(-1) = f(0) - f'(0) + \frac{1}{2}f''(0) - \frac{1}{6}f'''(\xi_2)$$

$$\frac{f(1) - f(-1)}{2} - f'(0) = \frac{1}{6} \cdot \frac{f'''(\xi_1) + f'''(\xi_2)}{2}$$

$f(x)$ 在 $[-1, 1]$ 上有三阶导数, 故 $f'''(x)$ 在 $[-1, 1]$ 上有介值性.

故 $\exists \xi \in [-1, 1]$, 使 $f'''(\xi) = \frac{f'''(\xi_1) + f'''(\xi_2)}{2}$.

$$\text{即 } \exists \xi \in [-1, 1], \text{ 使 } \frac{f'''(\xi)}{6} = \frac{f(1) - f(-1)}{2} - f'(0)$$

例. 设 $f(x) \in D^2[0, 1]$, $f(0) = f(1) = 1$, $\min_{0 \leq x \leq 1} f(x) = -1$,

证明: $\exists \xi \in (0, 1)$, 使 $f''(\xi) \geq 16$.

证. 由题意, $\exists x_0 \in (0, 1)$, 使 $f(x_0) = -1 = \min_{0 \leq x \leq 1} f(x)$. $f'(x_0) = 0$.

$\exists \xi_x \in [0, 1]$, 使

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(\xi_x)(x - x_0)^2$$

$$f(0) = -1 + \frac{1}{2}f''(\xi_0)x_0^2, \quad f''(\xi_0) = \frac{4}{x_0^2}$$

$$f(1) = -1 + \frac{1}{2}f''(\xi_1)(1 - x_0)^2, \quad f''(\xi_1) = \frac{4}{(1 - x_0)^2}$$

$$0 < x_0 \leq \frac{1}{2}, f''(\xi_0) = \frac{4}{x_0^3} \geq \frac{1}{16}, \text{取 } \xi = \xi_0 \text{ 即可.}$$

$$\frac{1}{2} < x_0 < 1, f''(\xi_1) = \frac{4}{(1-x_0)^2} \geq \frac{1}{16}, \text{取 } \xi = \xi_1 \text{ 即可.}$$

例. 求 $\lim_{x \rightarrow 0} \frac{\ln(1+x) \ln(1-x) - \ln(1-x^2)}{x^2 - \sin^2 x}$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) \ln(1-x) - \ln(1-x^2)}{x^2 - \sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + o(x^4)][-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + o(x^4)] - [-x^2 - \frac{x^4}{2} + o(x^4)]}{x^2 - (x - \frac{x^3}{6} + o(x^3))^2}$$

$$= \lim_{x \rightarrow 0} \frac{-x^2 - \frac{5}{12}x^4 + o(x^4) - x^2 + \frac{1}{2}x^4 + o(x^4)}{x^2 - [x^2 - \frac{1}{3}x^4 + o(x^4)]}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{12}x^4 + o(x^4)}{\frac{1}{3}x^4 + o(x^4)} = \frac{1}{4}$$

例. 求 $\lim_{x \rightarrow 0} \frac{x(\cos x - e^{-\frac{x^2}{2}})}{\tan x - x - \frac{x^3}{3}}$

$$\lim_{x \rightarrow 0} \frac{x(\cos x - e^{-\frac{x^2}{2}})}{\tan x - x - \frac{x^3}{3}} = \lim_{x \rightarrow 0} \frac{x(\cos x - e^{-\frac{x^2}{2}})}{\sin x - (x + \frac{x^3}{3}) \cos x} \cos x = \lim_{x \rightarrow 0} \frac{x(\cos x - e^{-\frac{x^2}{2}})}{\sin x - (x + \frac{x^3}{3}) \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{x[(1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)) - (1 - \frac{x^2}{2} + \frac{x^4}{8} + o(x^4))]}{x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5) - (x + \frac{x^3}{3})(1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4))}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{12}x^5 + o(x^5)}{\frac{2}{15}x^5 + o(x^5)} = -\frac{5}{8}$$

例. 求 $\lim_{n \rightarrow \infty} n \sin(2\pi e n!)$

$$e^x = 1 + x + \frac{x^2}{2} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!} + \frac{e^{\theta x}}{(n+2)!} x^{n+2}, \theta \in (0, 1).$$

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{(n+1)!} + \frac{e^\theta}{(n+2)!}, \theta \in (0, 1)$$

$$\begin{aligned} 2\pi e \cdot n! &= 2\pi \left[n! + \frac{n!}{1!} + \frac{n!}{2!} + \cdots + \frac{n!}{n!} + \frac{n!}{(n+1)!} + \frac{e^\theta n!}{(n+2)!} \right] \\ &= 2\pi \left[n! + n! + \frac{n!}{2!} + \cdots + 1 + \frac{1}{n+1} + \frac{e^\theta}{(n+2)(n+1)} \right] \end{aligned}$$

$$\lim_{n \rightarrow \infty} n \sin(2\pi e n!) = \lim_{n \rightarrow \infty} n \sin \left[2\pi \left(\frac{1}{n+1} + \frac{e^\theta}{(n+2)(n+1)} \right) \right]$$

$$= \lim_{n \rightarrow \infty} 2\pi \left(\frac{1}{1+\frac{1}{n}} + \frac{1}{1+\frac{1}{n}} \cdot \frac{e^\theta}{n+2} \right) = 2\pi.$$

4. 函数的极值与最值.

定义 称 $f'(x) = 0$ 的点为 $f(x)$ 的驻点.

极值点判定定理

第一充分条件 设 $f(x)$ 在 $U(x_0)$ 内连续, 在 $\dot{U}(x_0)$ 上可导, 在 $U(x_0)$ 内,

(i) 若 $x < x_0$, $f'(x) < 0$ 且 $x > x_0$, $f'(x) > 0$, 则 x_0 为 $f(x)$ 的一个极大值点.

(ii) 若 $x < x_0$, $f'(x) > 0$ 且 $x > x_0$, $f'(x) < 0$, 则 x_0 为 $f(x)$ 的一个极小值点.

第二充分条件 设 $f(x)$ 在 x_0 处有二阶导数, $f'(x_0) = 0$,

(i) 若 $f''(x_0) > 0$, 则 x_0 为 $f(x)$ 的极小值点,

(ii) 若 $f''(x_0) < 0$, 则 x_0 为 $f(x)$ 的极大值点.

例: 设 $y = f(x)$ 由 $2y^3 - 2y^2 + 2xy - x^2 = 1$ 确定, 求 $y = f(x)$ 的极值点.

$$\text{两边求导, } 6y^2 y' - 4y y' + 2xy' + 2y - 2x = 0, (*)$$

$$y' = \frac{x-y}{3y^2-2y+x} \quad \text{由} \begin{cases} y' = 0 \\ 2y^3-2y^2+2xy-x^2=1 \end{cases} \text{得} \begin{cases} x=1 \\ y=1 \end{cases}$$

(*) 式两边求导: $12y \cdot y' \cdot y' + 6y^2 \cdot y'' - 4y' \cdot y' - 4y \cdot y' + 2y' + 2y' + 2xy'' - 2 = 0$

$$y'' = \frac{(2-6y)(y')^2 - 2y' - 1}{3y^2 - 2y + x}, \quad y''|_{x=1} = \frac{1}{2} > 0.$$

$x=1$ 为 $y=f(x)$ 极小值点.

5 函数的凹凸性

定义 设 $f(x)$ 在 $[a, b]$ 上连续, 若 $\forall x_1, x_2 \in [a, b]$, 恒有 $f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1)+f(x_2)}{2}$, 则称 $f(x)$ 在 $[a, b]$ 上为凹函数.

设 $f(x)$ 在 $[a, b]$ 上连续, 若 $\forall x_1, x_2 \in [a, b]$, 恒有 $f\left(\frac{x_1+x_2}{2}\right) > \frac{f(x_1)+f(x_2)}{2}$, 则称 $f(x)$ 在 $[a, b]$ 上为凸函数.

定义 设 $f(x)$ 在 $[a, b]$ 上连续, 若 $\forall x_1, x_2 \in [a, b]$, $\forall \lambda \in (0, 1)$, $f(\lambda x_1 + (1-\lambda)x_2) < \lambda f(x_1) + (1-\lambda)f(x_2)$, 则称 $f(x)$ 在 $[a, b]$ 上为凹函数.

设 $f(x)$ 在 $[a, b]$ 上连续, 若 $\forall x_1, x_2 \in [a, b]$, $\forall \lambda \in (0, 1)$,

$f(\lambda x_1 + (1-\lambda)x_2) > \lambda f(x_1) + (1-\lambda)f(x_2)$, 则称 $f(x)$ 在 $[a, b]$ 上为凸函数.

定理 设 $f(x)$ 在 $[a, b]$ 上连续, 在 (a, b) 上可导, 则 $f(x)$ 在 $[a, b]$ 上为凹(凸)函数 $\Leftrightarrow f'(x)$ 单调递减(递增).

定理 设 $f(x)$ 在 $[a, b]$ 上连续, 在 (a, b) 上有二阶导数, 则 $f(x)$ 在 $[a, b]$ 上为凹(凸)函数 $\Leftrightarrow f''(x) \geq 0$ (≤ 0).

定义 设 $f(x)$ 在区间 I 上连续, 若函数图像在 $(x_0, f(x_0))$ 两侧的凹凸性相反, 则称 $(x_0, f(x_0))$ 为 $f(x)$ 的拐点.

定理 设 $(x_0, f(x_0))$ 为 $f(x)$ 的一个拐点, 且 $f'(x)$ 存在, 则 $f''(x_0) = 0$.

设 $f''(x_0) = 0, f'''(x_0) \neq 0$, 则 $(x_0, f(x_0))$ 为 $f(x)$ 的拐点.

定理 设 $f'(x_0) = f''(x_0) = \dots = f^{(n-1)}(x_0) = 0$, 且 $f^{(n)}(x_0) \neq 0$,

$n = 2k, k \in \mathbb{N}, x = x_0$ 为 $f(x)$ 的极值点, 且 $(x_0, f(x_0))$ 不为 $f(x)$ 的拐点.

$n = 2k+1, k \in \mathbb{N}, x = x_0$ 不为 $f(x)$ 的极值点, 而 $(x_0, f(x_0))$ 为 $f(x)$ 的拐点.

例 设 $x_i > 0, 1 \leq i \leq n$ 且 $\sum_{i=1}^n x_i = 1, a > 1$

证明:
$$\sum_{i=1}^n (x_i + \frac{1}{x_i})^a \geq \frac{(n^2+1)^a}{n^{a-1}}.$$

证. 令 $f(x) = (x + \frac{1}{x})^a, a > 1, x > 0$

$$f'(x) = a(x + \frac{1}{x})^{a-1} (1 - \frac{1}{x^2})$$

$$f''(x) = a(a-1)(x + \frac{1}{x})^{a-2} (1 - \frac{1}{x^2})^2 + a(x + \frac{1}{x})^{a-1} \cdot \frac{2}{x^3} > 0$$

$f(x)$ 在 $(0, +\infty)$ 上为凹函数

$$f(\frac{x_1 + x_2 + \dots + x_n}{n}) \leq \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}$$

$$\text{即 } \frac{\sum_{i=1}^n (x_i + \frac{1}{x_i})^a}{n} \geq (\frac{\sum_{i=1}^n x_i}{n} + \frac{1}{\frac{\sum_{i=1}^n x_i}{n}})^a$$

$$\text{又 } x_1 + x_2 + \dots + x_n = 1$$

$$\text{故 } \sum_{i=1}^n (x_i + \frac{1}{x_i})^a \geq n \cdot (\frac{1}{n} + n)^a = \frac{(n^2+1)^a}{n^{a-1}}$$

6 曲线的渐近线与曲率

定义 设曲线 $y=f(x)$ 上动点 $M(x, f(x))$ 沿自变量某一变化方向运动时, M 无限远离坐标原点, 且 M 到定直线 $l: Ax+By+C=0$ ($A^2+B^2 \neq 0$) 的距离趋于 0, 则称 l 为曲线 $y=f(x)$ 在该自变量变化方向下的渐近线.

1° $x \rightarrow x_0^+$ (或 x_0^-) 时

$$\begin{cases} \lim_{x \rightarrow x_0^+} f(x) = \infty \\ \lim_{x \rightarrow x_0^+} (Ax + Bf(x) + C) = 0 \end{cases} \Leftrightarrow \begin{cases} \lim_{x \rightarrow x_0^+} f(x) = \infty \\ B=0 \text{ 且 } Ax_0 + C = 0 \end{cases}$$

$$l: x = -\frac{C}{A} = x_0.$$

即当 $\lim_{x \rightarrow x_0^+} f(x) = \infty$ 时, $y=f(x)$ 以 $x=x_0$ 为渐近线, 称之为铅直渐近线.

2° $x \rightarrow +\infty$ (或 $x \rightarrow -\infty$) 时.

$$\lim_{x \rightarrow +\infty} (Ax + Bf(x) + C) = 0. \text{ 设 } Ax + Bf(x) + C = \alpha \quad (x \rightarrow +\infty, \alpha \rightarrow 0)$$

$$B \neq 0, f(x) = -\frac{A}{B}x - \frac{C}{B} + \frac{1}{B}\alpha, \text{ 则 } l: y = -\frac{A}{B}x - \frac{C}{B}$$

$$\text{设 } l: y = kx + b$$

$$k = \lim_{x \rightarrow +\infty} \frac{f(x)}{x}$$

$$b = \lim_{x \rightarrow +\infty} (f(x) - kx)$$

例 求 $f(x) = \frac{x^2 \arctan x}{x-1}$ $-x$ 的渐近线.

$\lim_{x \rightarrow 1} f(x) = \infty$, $x=1$ 为 $f(x)$ 的铅直渐近线.

$$k = \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \frac{x \arctan x}{x-1} - 1 = \lim_{x \rightarrow +\infty} \frac{\arctan x}{1 - \frac{1}{x}} - 1 = \frac{\pi}{2} - 1$$

$$b = \lim_{x \rightarrow +\infty} (f(x) - kx) = \lim_{x \rightarrow +\infty} \left[\frac{x^2 \arctan x}{x-1} - x - \left(\frac{\pi}{2} - 1\right)x \right] = \lim_{x \rightarrow +\infty} \left(\frac{x^2 \arctan x}{x-1} - \frac{\pi}{2}x \right)$$

$$= \lim_{x \rightarrow +\infty} \frac{x^2 \arctan x - \frac{\pi}{2}x(x-1)}{x-1} = \lim_{x \rightarrow +\infty} \frac{\arctan x - \frac{\pi}{2}(1 - \frac{1}{x})}{\frac{1}{x} - \frac{1}{x^2}}$$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{1}{1+x^2} - \frac{\pi}{2} \cdot \frac{1}{x^2}}{-\frac{1}{x^2} + \frac{2}{x^3}} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{1+\frac{1}{x^2}} - \frac{\pi}{2}}{-1 + \frac{2}{x}} = \frac{\pi}{2} - 1$$

同理 $x \rightarrow -\infty$, $k = -\frac{\pi}{2} - 1$, $b = -\frac{\pi}{2} - 1$

$y = (\frac{\pi}{2} - 1)(x-1)$ 和 $y = -(\frac{\pi}{2} + 1)(x-1)$ 为 $f(x)$ 的斜渐近线

定义 设曲线 $L: y = f(x)$, ($a \leq x \leq b$), 所有“内接折线”的长度有界, 则称 L 为可求长曲线, 称 L 的“内接折线”长度上确界为 L 的长度.

引理 设 $y = f(x)$ 在 $[a, b]$ 上有一阶连续导数, 则 $L: y = f(x)$ 在 $[a, b]$ 可求长

$$\text{此时 } \lim_{n^* \rightarrow n} \frac{\widehat{mn}^*}{|mn^*|} = 1.$$

定理 设 $y = f(x)$ 在 $[a, b]$ 上有一阶连续导数, 记 $s = \widehat{mn}^*$, 则

$$\text{弧长微元 } ds = \sqrt{1 + (f'(x))^2} dx$$

$$\text{有 } ds^2 = dx^2 + dy^2$$

$$\text{若 } \begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \text{ 则 } ds = \sqrt{(\varphi'(t))^2 + (\psi'(t))^2}$$

定义 记 $\widehat{mn}^*$ 两端点切线的转角为 $\Delta\alpha$, 称 $\lim_{\Delta s \rightarrow 0} \frac{\Delta\alpha}{\Delta s}$ 为曲线 $y = f(x)$ 在 M 点处的曲率, 记为 $k = \left| \frac{d\alpha}{ds} \right| = \lim_{\Delta s \rightarrow 0} \frac{\Delta\alpha}{\Delta s}$.

由导数的几何意义, $y' = \tan \alpha$. $\alpha = \arctan y'$

$$d\alpha = \frac{1}{1+(y')^2} dy', \quad dy' = y'' dx, \quad d\alpha = \frac{y''}{1+(y')^2} dx$$

$$ds = \sqrt{1+(y')^2} dx,$$

$$k = \left| \frac{d\alpha}{ds} \right| = \left| \frac{y''}{(1+(y')^2)^{\frac{3}{2}}} \right|.$$

若 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$, 则 $k = \left| \frac{\psi''(t)\varphi'(t) - \psi'(t)\varphi''(t)}{[(\varphi'(t))^2 + (\psi'(t))^2]^{\frac{3}{2}}} \right|$

定义 称曲线 L 在点 M 处的曲率 k_M 的倒数 $\frac{1}{k_M}$ 为曲线 L 在点 M 处的曲率半径, 记为 R_M .

称曲线 L 在点 M 处有相同切线, 相同凹凸方向, 相同弯曲程度的圆 C 为曲线 L 在点 M 处的曲率圆. 称圆 C 的圆心 (ξ, η) 为曲线 L 在点 M 处的曲率中心.

$$\text{其中 } \xi = x - \frac{y'(1+(y')^2)}{y''}, \quad \eta = y + \frac{1+(y')^2}{y''}$$

例. 设 $f(x)$ 在 $[a, b]$ 上连续, 在 (a, b) 上二次可导, 证明:

$$\exists \xi \in (a, b), \text{ 使 } \frac{f(x) - f(a)}{x - a} - \frac{f(b) - f(a)}{b - a} = \frac{1}{2}(x - b)f''(\xi)$$

证. 一.

$$\text{令 } g(x) = \frac{f(x) - f(a)}{x - a}, \quad g(x) \in C[a, b], \quad g(x) \in D(a, b),$$

由拉格朗日中值定理, $\exists \eta \in (x, b) \subseteq (a, b)$.

$$\text{使 } g(x) - g(b) = g'(\eta) \cdot (x - b)$$

$$\text{其中 } g'(\eta) = \frac{f'(\eta)(\eta - a) - (f(\eta) - f(a))}{(\eta - a)^2} = \frac{f(a) - f(\eta) - f'(\eta)(a - \eta)}{(\eta - a)^2}$$

由泰勒中值定理, $\exists \xi \in (a, \eta) \subseteq (a, b)$

$$\text{使 } f(a) = f(\eta) + f'(\eta)(a-\eta) + \frac{1}{2}f''(\xi)(a-\eta)^2$$

$$\text{故 } f(a) - f(\eta) - f'(\eta)(a-\eta) = \frac{1}{2}f''(\xi)(a-\eta)^2$$

$$\text{从而 } g(x) - g(b) = \frac{1}{2}(x-b)f''(\xi)$$

$$\text{即 } \frac{f(x) - f(a)}{x-a} - \frac{f(b) - f(a)}{b-a} = \frac{1}{2}(x-b)f''(\xi)$$

$$\text{二. } \quad \text{记 } z = \frac{\frac{f(x) - f(a)}{x-a} - \frac{f(b) - f(a)}{b-a}}{x-b} = A$$

$$\text{即 } z[(b-a)(f(x) - f(a)) - (x-a)(f(b) - f(a))] - A(x-a)(x-b)(b-a) = 0$$

$$\text{令 } g(t) = z[(b-a)(f(t) - f(a)) - (t-a)(f(b) - f(a))] - A(t-a)(t-b)(b-a)$$

$$t \in [a, b], \text{ 则 } g(a) = 0, g(b) = 0, g(x) = 0.$$

$$g(t) \in C[a, b], g(t) \in D(a, b).$$

$$\text{由罗尔中值定理, } \exists a < \eta_1 < x < \eta_2 < b, \text{ 使 } g'(\eta_1) = g'(\eta_2) = 0.$$

$$\text{而 } g'(t) \in C[a, b], g'(t) \in D(a, b).$$

$$\text{由罗尔中值定理, } \exists \xi \in (\eta_1, \eta_2), \text{ 使 } g''(\xi) = 0$$

$$\text{而 } g''(\xi) = z(b-a)(f''(\xi) - A) = 0. \text{ 故 } f''(\xi) = A.$$

$$\text{即 } \frac{f(x) - f(a)}{x-a} - \frac{f(b) - f(a)}{b-a} = \frac{1}{2}(x-b)f''(\xi).$$

$$\text{例. 求 } \lim_{x \rightarrow +\infty} x^3 \left(\arctan \frac{3x^3 + x}{x^2 + 1} - \arctan \frac{3x^3 - x}{x^2 + 1} \right)$$

$$\text{由拉格朗日中值定理, } x > 0, \exists \xi \in \left(\frac{3x^3 - x}{x^2 + 1}, \frac{3x^3 + x}{x^2 + 1} \right)$$

$$\text{使 } \arctan \frac{3x^3+x}{x^2+1} - \arctan \frac{3x^3-x}{x^2+1} = (\arctan \xi)' \left(\frac{3x^3+x}{x^2+1} - \frac{3x^3-x}{x^2+1} \right) \\ = \frac{1}{1+\xi^2} \cdot \frac{2x}{x^2+1}$$

$$\text{当 } x \rightarrow +\infty, \frac{3x^2+1}{x^2+1} \rightarrow 3, \frac{3x^2-1}{x^2+1} \rightarrow 3, \text{ 故 } \frac{\xi}{x} \rightarrow 3$$

$$\lim_{x \rightarrow +\infty} x^3 \left(\arctan \frac{3x^3+x}{x^2+1} - \arctan \frac{3x^3-x}{x^2+1} \right)$$

$$= \lim_{x \rightarrow +\infty} \frac{1}{1+\xi^2} \cdot \frac{2x^4}{x^2+1} = \lim_{x \rightarrow +\infty} \frac{1}{\frac{1}{x^2} + \left(\frac{\xi}{x}\right)^2} \cdot \frac{2}{1+\frac{1}{x^2}} = \frac{2}{9}.$$